IAS PRELIMS/MAINS

Ecology & Environment

A COMPREHENSIVE COVERAGE

MODULE - 30

WATER POLLUTION - 7

GEOGENIC CONTAMINANTS OF GROUND WATER

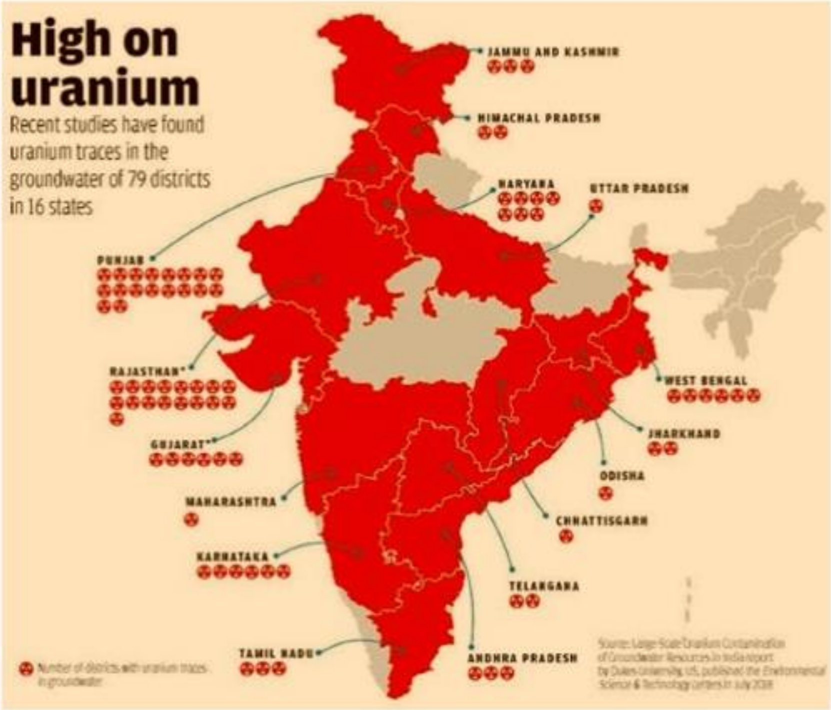
URANIUM

- According to a 2018 study, aquifers in 16 states of India are contaminated by the presence of Uranium.
- **The study has found over 30mcg/l of uranium in groundwater in parts of North-western, Southern and South-eastern India.**
- Drinking such water can severely damage kidneys; **the WHO prescribes 30mcg/l as the upper limit.**
- Data from 324 wells in Rajasthan and Gujarat shows prevalence of high uranium concentrations.
- Unfortunately the residents of the areas surveyed were using the contaminated water as their drinking water.
- **These findings highlight a major gap in India's water quality monitoring. As the Bureau of Indian Standards does not specify a norm for uranium level, water is not tested regularly for it.**

Learning Space

URANIUM

- Occurrence of uranium in groundwater depends on many factors
  - ✓ The amount of uranium contained in aquifer's rocks,
  - ✓ Water-rock interactions that cause uranium to be extracted from the rocks,
  - ✓ Oxidation conditions that enhance the extracted uranium's solubility in water and
  - ✓ The interaction of uranium with chemicals and contaminants in the water which can further enhance its solubility.



Conclusion

So, even though the presence of uranium in groundwater is natural, groundwater depletion due to excessive withdrawal of water for irrigation and nitrate pollution of groundwater due to excessive use of fertilizers maybe aggravating the problem according to the study (enhanced uranium mobilization).

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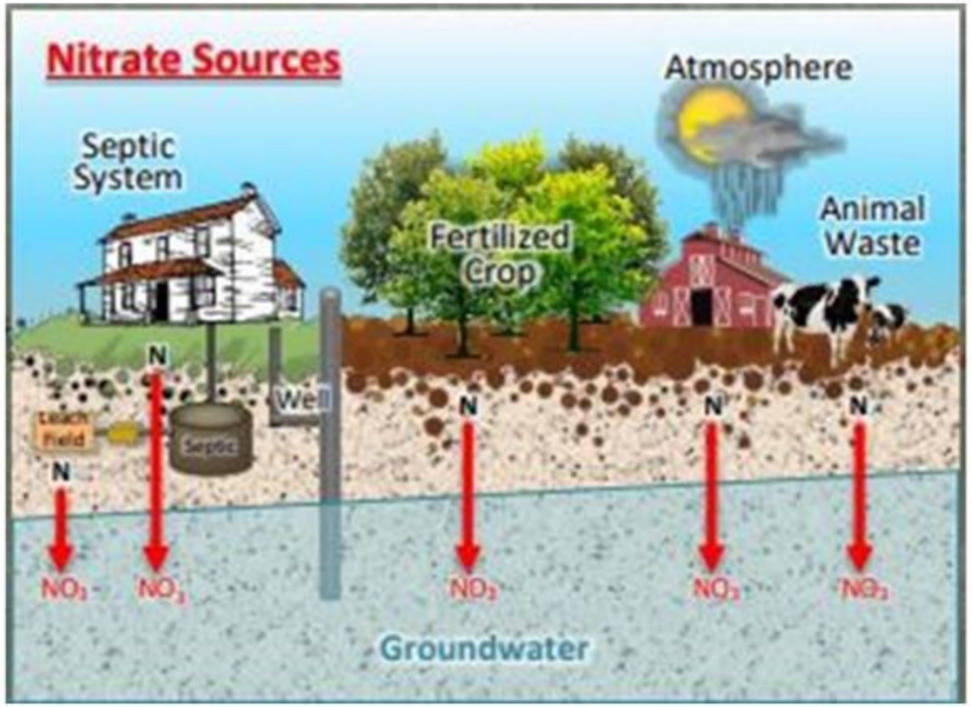
ANTHROPOGENIC CONTAMINANTS OF GROUND WATER

NITRATES

- Elevated concentrations (more than 45PPM) of nitrates is the most widespread among anthropogenic contaminants.
- The concentrations are increasing with increasing use of nitrogenous fertilizers and disposal of untreated domestic sewage.
- Such contamination has been reported from parts of 21 states.



GROUNDWATER CONTAMINANTS AND THEIR EFFECTS

1 NITRATES

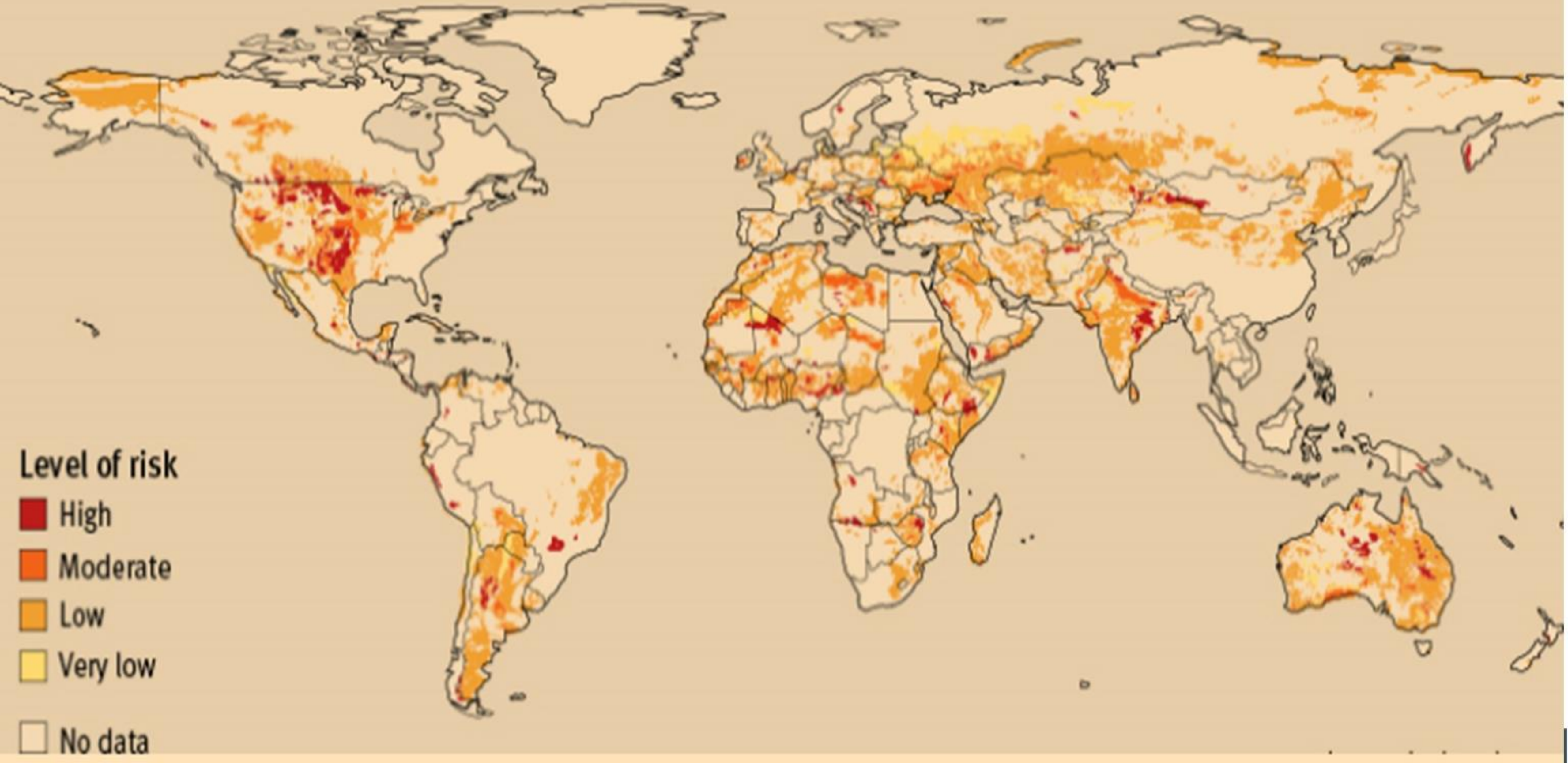
- Excess nitrates in drinking water react with hemoglobin to form Methemoglobin, which stops Oxygen transfer to different cells and tissues.
- This condition is called Methemoglobinemia or Blue baby Syndrome.
- **Blue-Baby Syndrome** - It causes Cyanosis and Cyanotic heart defect in infants.
- Cyanosis is defined as the bluish or purplish discoloration of the skin and mucous membranes due to the tissues having low oxygen concentration.
- Cyanotic heart defect occurs due to un-oxygenated blood entering into circulation.
- **Nitrates Problem in India**
  - ✓ In states such as Punjab, Haryana, Delhi, Maharashtra, and Andhra Pradesh with intense economic and agricultural activities, nitrate concentrations are increasing in ground and surface waters.

GROUNDWATER CONTAMINANTS AND THEIR EFFECTS

2 ARSENIC

- It is a highly toxic pollutant. Long term exposure to arsenic causes Black Foot disease and skin cancer.
- **Black-foot Disease** - It is a severe disease where the blood vessels in the lower limbs are severely damaged which results in Gangrene.
- Gangrene is tissue death caused by not enough supply of blood.
- Seepage of industrial and mine discharges can lead to arsenic deposition in ground water.
- **Arsenic Problem in India**
  - ✓ In India and Bangladesh in Ganges Delta, millions of people are exposed to ground water contaminated with high levels of arsenic which is naturally occurring in high concentrations deeper levels of groundwater.

Estimated Risk of Arsenic in Drinking Water



Learning Space

GROUNDWATER CONTAMINANTS AND THEIR EFFECTS

3 FLUORIDE

- Excess fluoride in drinking water causes Fluorosis disorder whose symptoms are teeth deformity, painful joints and outward bending of legs from knees called Knock-Knee syndrome.

- **Fluoride problem in India**
  - ✓ Lack of clean drinking water has put over 12 million people in India at a very high risk of fluorosis.
  - ✓ The Ministry of Health & Family Welfare has identified at least 132 districts in 19 states severely affected by high fluoride content in drinking water, a leading cause of fluorosis.

GROUNDWATER CONTAMINANTS AND THEIR EFFECTS

4 MERCURY

- Mercury compounds in water are converted into toxic methyl mercury by bacteria, it causes a severe neurological disease in humans known as Minamata which causes paralysis and mental derangement.
- Symptoms include numbness in the hands and feet, general muscle weakness, narrowing of the field of vision and damage to hearing and speech.
- In extreme cases, insanity, paralysis, coma and death occur.

- **Lead and Cadmium problem in India**
  - ✓ Lead poisoning causes anemia, damage to nervous system, damage to liver and kidneys, muscle paralysis and cancer.
  - ✓ Cadmium poisoning causes cancer of liver also causes a disease called Itai-Itai which is a painful disease of bones and joints and also causes softening of bones.

WATER POLLUTION EXTENT IN INDIA

- India is fast running out of water due to pollution, over-extraction and climate change.
- A 2015 report by WaterAid says that 80% of the surface water in the country is polluted.
- Domestic sewage, inadequate sanitation facilities and poor sewage treatment facilities are responsible for the growing pollution.
- The rivers, the main source of surface water are drying. Other than pollution the rivers are being affected by diversion of flows, loss of biodiversity, sand mining etc.

WaterAid

WATER POLLUTION EXTENT IN INDIA

- Water bodies like ponds and lakes have also been either encroached upon or have become dumping grounds of sewage and waste.
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- The CPCB in association with State Pollution Control Boards (SPCBs) is monitoring the quality of water bodies at 2500 locations across the country under **National Water Quality Monitoring Programme** (NWQMP).
- **The results from the programme indicate that organic pollution is the predominant cause of water pollution.**

PARAMETERS MEASURED UNDER NATIONAL WATER QUALITY MONITORING PROGRAMME

FIELD OBSERVATIONS (7)

- Weather
- Depth of main stream/death of water table
- Colour and intensity
- Odour
- Visible effluent discharge
- Station detail

CORE PARAMETERS (9)

- pH
- Temperature
- Conductivity, μmhos/cm
- Dissolved Oxygen, mg/L
- BOD, mg/L
- Nitrate – N, mg/L
- Faecal Caliform, MPN/100 ml
- Total Coliform, MPN/100 ml

CORE PARAMETERS (19)

- Turbidity, NTU
- Phenolphthalein Alkalinity, as $CACO_3$
- Chlorides, mg/L
- Total Kjeldahl –N, as N mg/L
- Ammonia- N as N mg/L
- Hardness, as $CaCO_3$
- Calcium, as $CaCO_3$
- Suphate, mg/L
- Sodium, mg/L
- Total Dissolved Solids, mg/L
- Total Fixed Dissolved Solids, mg/L
- Total Suspended Solid, mg/L
- Phosphate, mg/L
- Boron, mg/L
- Magnesium, as $CaCO_3$
- Potassium, mg/L
- Fluoride, mg/L

BIO-MONITORING (3)

- Saprobity Index
- Diversity Index
- P/R Ratio

TRACE METALS (9)

- Arsenic, μg/L
- Cadmium, μ g/L
- Copper, μ g/L
- Lead μ g/L
- Chromium (Total) μ g/L
- Nickel, μ g/L
- Zinc, μ g/L
- Mercury, μ g/L
- Iron (Total), μ g/L

PESTICIDES (15)

- Alpha BHC, μg/L
- Beta BHC, μg/L
- Gama BHC (Lindane) , μg/L
- O P DDT, μg/L
- P P DDT, μg/L
- Alpha Endosulphan, μg/L
- Beta Endosulphan, μg/L
- Aldrin, μg/L
- Dieldrin, μg/L
- Carboryl (Carbamate) , μg/L
- 2-4D, μg/L
- Malathian, μg/L
- Methyl Parathian, μg/L
- Anilophos, μg/L
- Chloropyrphos, μg/L

Learning Space

2018 REPORT OF CPCB

- Of India's states and UTs, 27 have critically polluted river stretches according to the 2018 report by CPCB.
- The report identified 351 polluted stretches on 323 rivers in states and UTs. Maharashtra has the highest 53 polluted river stretches followed by Assam, MP, Kerala and Gujarat.
- In some river stretches the quality of water is so poor that it requires immediate intervention. CPCB puts a water body under this category when its BOD exceeds 30mg/l.
- The number of such critically polluted stretches has increased across the country from 32 to 45 in the last three years.
- Vasishta Nadhi (TN) is officially the dirtiest river in the country, being ranked among the 351 polluted stretches studied by the CPCB in 2018. The Biochemical Oxygen Demand (BOD) of 500mg/l in the river is 200 times more than the standard limit.

CPCB DESIGNATED BEST USE

- CPCB has developed a concept called 'Designated Best use' .
- The water quality is designated as A, B, C, D and E on the basis of
  - ✓ PH
  - ✓ Do-mg/l
  - ✓ BOD-mg/l
  - ✓ Total coliform
  - ✓ Free ammonia mg/l
  - ✓ Electrical conductivity

DESIGNATED –BEST –USE	CLASS OF WATER	CRITERIA
Drinking water source without conventional treatment but after disinfection	A	▪ Total coliforms organism MPN/100 ml shall be 0 or less ▪ PH between 6.5 and 8.5 ▪ Dissolved oxygen: 6 mg/l or more Biological oxygen demand 5 days 20c mg/l or less
Outdoor bathing	B	▪ Total coliforms organism MPN/100 ml shall be 00 or less ▪ PH between 6.5 and 8.5 ▪ Dissolved oxygen: 5 mg/l or more ▪ Biological oxygen demand 5 days 20c 3mg/l or less
Drinking water source after conventional treatment and disinfection	C	▪ Total coliforms organism MPN/100 ml shall be 00 or less ▪ PH between 6 to 9 ▪ Dissolved oxygen: 4 mg/l or more ▪ Biological oxygen demand 5 days 20c mg/l or less
Propagation of wild life and fisheries	D	▪ PH between 6.5 to 8.5 ▪ Dissolved oxygen: 4 mg/l or more ▪ Free Ammonia(as N): 1.2 mg/l
Imigation, Industrial cooling, controlled waste disposal	E	▪ PH between 6.0 to 8.5 ▪ Electrical conductivity at 25c micro mhos/cm max 2250 ▪ Sodium absorption ratio max 26 ▪ Boron max 2 mg/l ▪ Below E-Not meeting A ,B, C, D, and E criteria

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SINCERE THANKS
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